

# WEVSR: Video Diffusion Generators for Real-World Video Super-Resolution with Wavelet-Enhanced VAE Encoder

Yuying Chen<sup>1,2</sup> Zhirui Liu<sup>2,3</sup> Linyan Jiang<sup>3</sup> Qifan Gao<sup>3</sup> Xianguo Zhang<sup>3</sup> Jianhou Gan<sup>4</sup> Wenqi Ren<sup>2,5</sup>

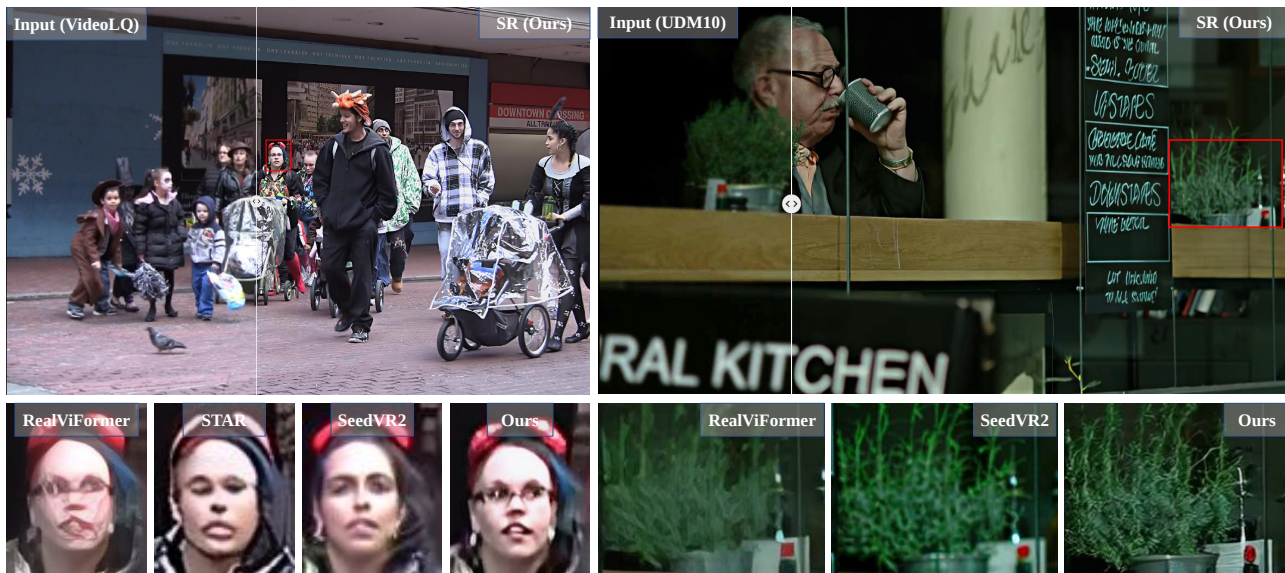


Figure 1. Visualization comparisons on both real-world and synthetic low-resolution videos. Compared to the state-of-the-art VSR methods RealViFormer (Zhang & Yao, 2024), STAR (Xie et al., 2025b), SeedVR2 (Wang et al., 2025a), our results demonstrate more natural facial details and better structure of the plants. (Zoom-in for best view)

## Abstract

Recent advances in video diffusion models have demonstrated remarkable generative capability, yet adapting these large pretrained text-to-video (T2V) models to video super-resolution (VSR) typically encounters challenges, such as artifacts introduced by complex degradations in real-world scenarios and compromised fidelity due to the strong generative capacity of the powerful T2V models. We present WEVSR, a novel approach that adapts a pretrained flow-matching video diffusion transformer to VSR. First, we

design a task-oriented adaptation strategy that leverages timestep sampling and noise augmentation to enhance detail restoration while preserving structural stability. Second, we propose a lightweight multi-level discrete wavelet transform (DWT) front-end for the VAE encoder, injecting explicit frequency priors into the latent space without modifying the pretrained decoder. Extensive experiments across multiple VSR benchmarks demonstrate that WEVSR achieves state-of-the-art performance against existing approaches. Code and models will be released [here](#).

<sup>1</sup>Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen, China <sup>2</sup>Shenzhen Campus of Sun Yat-sen University, Shenzhen, China <sup>3</sup>Tencent Technology, Shenzhen, China <sup>4</sup>Yunnan Normal University, Kunming, China <sup>5</sup>Key Laboratory of Information Technology (Sun Yat-sen University), Ministry of Education, Guangzhou, China. Correspondence to: Wenqi Ren <renwq3@mail.sysu.edu.cn>.

## 1. Introduction

Real-world video super-resolution (RealVSR) aims to reconstruct high-resolution (HR) and temporally stable videos from low-resolution (LR) inputs suffering from complex degradations such as compression artifacts, motion blur, noise, and aliasing (Wang et al., 2021; Yang et al., 2021; Chan et al., 2022b; Zhang & Yao, 2024; Zhou et al., 2024a).

Despite the rapid progress of synthetic VSR benchmarks, RealVSR remains fundamentally challenging because the complex degradation process is unknown, spatial-temporal correlations are easily disrupted, and high-frequency details are often irreversibly lost. Recent diffusion- and transformer-based approaches attempt to address these issues by enhancing temporal modeling or introducing generative priors, yet their performance is still constrained by limited data diversity, insufficient generative capacity, or weak frequency modeling under real-world scenarios.

In parallel, large-scale video generative models, especially diffusion transformers (Peebles & Xie, 2023; Esser et al., 2024b) trained with flow-matching objectives (Lipman et al., 2022; Liu et al., 2022), have demonstrated remarkable ability to synthesize high-quality, temporally coherent videos (HaCohen et al., 2024; Wan et al., 2025). These models learn rich spatial-temporal priors from massive video corpora, capturing natural motion patterns, structural regularities, and fine-grained textures far beyond what conventional VSR models can access. This raises an intriguing question: How to adapt a powerful video generative model to RealVSR, without retraining it from scratch or compromising its generative priors? The generative training objective differs fundamentally from the restoration objective, since the VAE encoder-decoder is optimized for synthesis rather than reconstruction, and real-world degradations often break the assumptions embedded in generative priors.

To address these challenges, we propose a wavelet-enhanced video super-resolution (WEVSR) approach, adapting a pre-trained flow-based video diffusion transformer (DiT) for RealVSR. Our approach is built upon two key insights. First, the flow-matching process can be guided toward restoration-friendly behavior by modifying the noise-timestep dynamics. We design a task-oriented adaptation strategy that incorporates detail-oriented timestep sampling strategy and flow-based noise augmentation, enabling the model to better preserve global structures while enhancing robustness to real-world degradations. These mechanisms reshape the flow trajectory in a way that improves detail recovery without destabilizing temporal consistency.

Furthermore, the VAE encoder used in video generative models is inherently limited in capturing high-frequency information (Lai et al., 2025), as it is optimized for perceptual synthesis rather than precise reconstruction. To compensate for this limitation, we introduce a multi-level discrete wavelet transform (DWT) front-end before the VAE encoder. This module injects explicit frequency decomposition into the latent representation while maintaining full compatibility with the pretrained decoder. By exposing the encoder to multi-scale frequency cues, the diffusion transformer backbone gains a stronger ability to reconstruct textures, edges, and aliasing patterns under complex degradations.

Extensive experiments on various benchmarks verify that WEVSR achieves the state-of-the-art performance in both perceptual and fidelity quality. The qualitative comparison in Figure 1 show that our method produces sharper details, fewer artifacts, and more consistent motion across frames.

Our contributions are summarized as follows:

- We introduce the multi-level DWT for pixel-to-latent transfer and propose a wavelet-enhanced VAE encoder architecture, injecting frequency priors while keeping the latent space unchanged.
- We adapt a pretrained flow-based video DiT to RealVSR with a fidelity-aware condition controller to enhance detail recovery and structural stability.
- Extensive experiment demonstrate the state-of-the-art performance of our proposed approach on multiple RealVSR benchmarks.

## 2. Related Works

**Video super-resolution (VSR).** Video super-resolution has evolved from early CNN- and RNN-based architectures to more advanced transformer-based frameworks. Classical VSR methods (Caballero et al., 2017; Jo et al., 2018; Wang et al., 2019; Li et al., 2020; Chan et al., 2021; 2022a) focused on motion compensation, optical-flow-based alignment, and multi-frame fusion, but their performance drops significantly under real-world degradations where blur, noise, compression, and aliasing interact unpredictably. Transformer-based VSR (Liang et al., 2022; Zhao et al., 2022; Zhou et al., 2024b; Xu et al., 2024) improves long-range temporal modeling and motion reasoning, yet still struggles with unstable temporal correlations. Recent RealVSR works attempt to address these issues through degradation-aware modeling, temporal consistency constraints, or large-scale data augmentation, but their generative capacity remains limited by the scale and diversity of available training data. As highlighted by TurboVSR (Wang et al., 2025c), even strong architectures often fail to generalize when facing complex, mixed degradations, motivating the need for stronger priors and more robust restoration paradigms.

**Video Synthesis.** Large-scale video generative models have recently achieved remarkable progress, particularly diffusion transformers trained with flow-matching objectives. These models learn powerful spatial-temporal priors from massive video corpora, capturing natural motion dynamics, structural regularities, and fine-grained textures. Frameworks like WAN (Wan et al., 2025) and LTX-Video (HaCohen et al., 2024) demonstrate strong capabilities in synthesizing high-quality, temporally coherent videos. Their learned priors offer a promising direction for video restoration, as they encode rich distributions of natural motion and

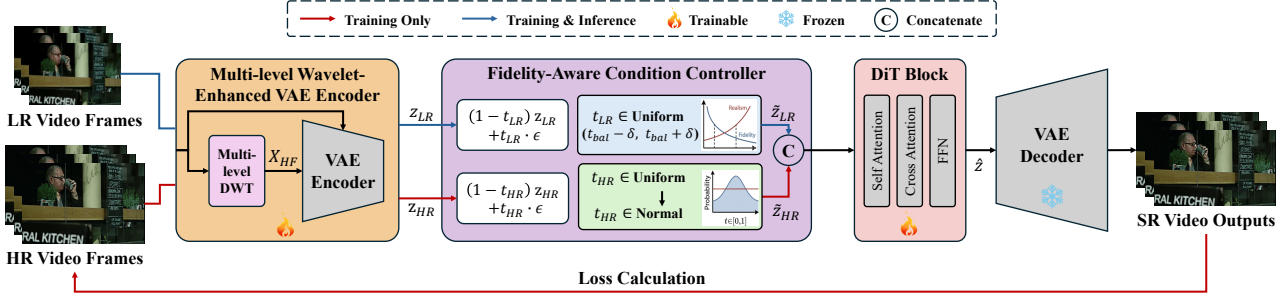


Figure 2. Overview of the proposed WEVSR framework. The input video frames are transferred to the latent variables by the multi-level wavelet-enhanced VAE encoder. The fidelity-aware condition controller then augments the latents via adding flow-based noise with fidelity-aware timestep sampling. At last, the VAE decoder reconstructs the denoise latents to high-quality videos.

texture. However, directly applying video generative models to VSR is challenging due to mismatched training objectives, limited reconstruction ability of VAE encoders, and the difficulty of handling real-world degradations without destabilizing temporal consistency.

**Diffusion-based VSR.** Existing diffusion-based VSR methods introduce temporal consistency modules, degradation-aware noise modeling, or multi-stage refinement strategies to stabilize the denoising trajectory (Zhou et al., 2024a; Xie et al., 2025b; Wang et al., 2025b; Li et al., 2025; Sun et al., 2025; Chen et al., 2025). While these approaches demonstrate promising results, they often require heavy retraining, rely on synthetic degradation assumptions, or operate on limited-scale VSR datasets, restricting their generative priors. Recent works (Zhuang et al., 2025; Xie et al., 2025a; Wang et al., 2025a) explore streaming-friendly diffusion, latent-cascaded restoration, or adversarial post-training, yet the gap between video generation and VSR remains substantial. In particular, current diffusion-based VSR methods rarely exploit the full potential of pretrained video DiTs.

### 3. Method

Our goal is to adapt a pretrained flow-based video diffusion transformer (DiT) to RealVSR in a training-efficient and degradation-robust manner. The proposed WEVSR framework consists of three key components: (1) a multi-level wavelet-enhanced VAE encoder that exploits frequency priors for pixel-to-latent compression, (2) a fidelity-aware condition controller that enables task-oriented adaptation, and (3) a video restoration pipeline that leverages the pretrained video DiT while keeping most of its generative priors. The overall architecture is illustrated in Figure 2. We use WAN-2.1 as the pretrained video generator.

#### Multi-level Wavelet-Enhanced VAE Encoder

Prior work such as FreqWarm (Lai et al., 2025) shows that decoders rely heavily on high-frequency latent signals to reconstruct fine details, while encoders tend to suppress

these high-frequency components, leading to limited exposure and systematic underfitting of high-frequency bands. To mitigate this mismatch without altering the pretrained decoder, we apply the multi-level discrete wavelet transform (DWT) only to the VAE encoder.

Given input video frames  $X$ , we apply DWT to each frame, producing multi-scale subbands:

$$X_{LL}, X_{LH}, X_{HL}, X_{HH} = \text{DWT}(X). \quad (1)$$

Among these subbands,  $X_{LL}$  mainly captures structural content while  $X_{LH}, X_{HL}, X_{HH}$  (jointly denoted by  $X_{HF}$ ) maintain most high-frequency information from the input. As shown in Figure 3, the  $X_{LL}$  subband will enter the next level DWT for further high-frequency information extraction, and the remaining subbands are injected into the original encoder branch to improve the exposure of high-frequency components. Note that each downsample layer of the encoder halves the height and width of the features, which naturally aligns with the inherent size reduction of the wavelet transform. Hence, we can direct concatenate the high-frequency subbands and the downsampled features channel-wisely, and apply a  $1 \times 1$  convolution to projecting the concatenated representation back to original channels:

$$X'_d = \text{Conv}_{1 \times 1}(\text{Concat}(X_d, X_{LH}, X_{HL}, X_{HH})), \quad (2)$$

where  $X_d$  is the downsampled features and  $X'_d$  is the fused features. This design ensures that the pretrained VAE encoder receives an input with enriched frequency cues while mostly preserving its original architecture. The encoder then produces a latent representation  $z$  that contains both structural and frequency-aware information.

#### Fidelity-Aware Condition Controller

We build upon a large-scale video diffusion transformer pretrained by a flow-matching paradigm. The VAE encoder first transfers the input frames into the latent representation, then the pretrained DiT models the probability flow between clean and noisy latent variables, and finally the VAE encoder restore the latents back to the pixel space.



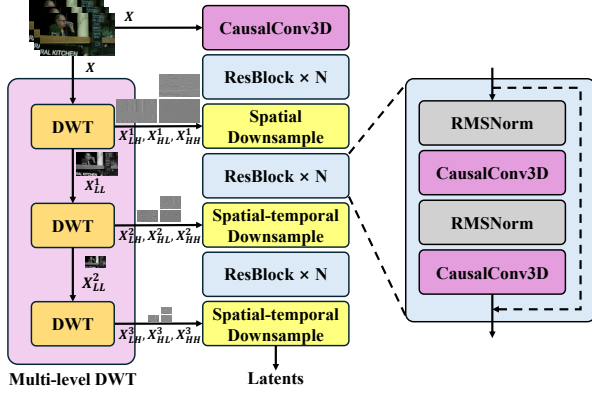


Figure 3. Architecture of the multi-level wavelet-enhanced VAE encoder. After each DWT, the high-frequency subbands are fused with downsampled features while the low-frequency subband enters the next level DWT for further high-frequency extractions. The usage of CasualConv3D (Yu et al., 2024) and RMSNorm (Zhang & Sennrich, 2019) follows WAN-2.1 (Wan et al., 2025).

Specifically, given the input low-resolution video frames  $X_{LR}$  and the target high-resolution video frames  $X_{HR}$ , we first derive the corresponding latent variables  $z_{LR}$  and  $z_{HR}$ . Subsequently, we apply flow-matching-based noise injection to obtain the noisy HR latent variable  $\tilde{z}_{HR}$  and the degraded LR latent variable  $\tilde{z}_{LR}$ :

$$\tilde{z}_{HR} = (1 - t_{HR})z_{HR} + t_{HR} \cdot \epsilon_{HR}, \quad (3)$$

$$\tilde{z}_{LR} = (1 - t_{LR})z_{LR} + t_{LR} \cdot \epsilon_{LR}, \quad (4)$$

where  $t_{HR}$  and  $t_{LR}$  are timesteps,  $\epsilon_{HR}$  and  $\epsilon_{LR}$  are standard Gaussian noises for HR and LR, respectively. Then, we concatenate  $\tilde{z}_{HR}$  and  $\tilde{z}_{LR}$  on channel dimension to feed into the DiT blocks to exploit the pretrained generative prior.

To stabilize the flow matching trajectory under real-world degradations, we introduce two mechanisms: *flow-based noise augmentation* and *fidelity-aware timestep sampling*. Flow-based noise augmentation injects degradation-aware perturbations into the latent space, encouraging the model to learn robust mappings that recover fine details even when the LR input contains severe aliasing or compression artifacts. Instead of uniformly sampling timestep as in common generative training, we bias the timestep sampling distribution toward the normal distribution for  $z_{HR}$ , selecting the timestep in the central region with greater probability, which empirically improves structural preservation, reduces over-smoothing, and concentrates on the detail restoration (Esser et al., 2024a). We further find the trade-off between fidelity and generation ability can be adjusted by the noising timestep of LR, as shown in Figure 4. For  $z_{LR}$ , the timestep is sampled uniformly from a chosen interval, determined by finding the intersection  $t_{bal}$  between the PSNR and CLIP-IQA curve drawn on a subset of the training dataset, with the half width  $\delta = 0.1$ . These mechanisms reshape the flow-matching dynamics to better suit the restoration objective.

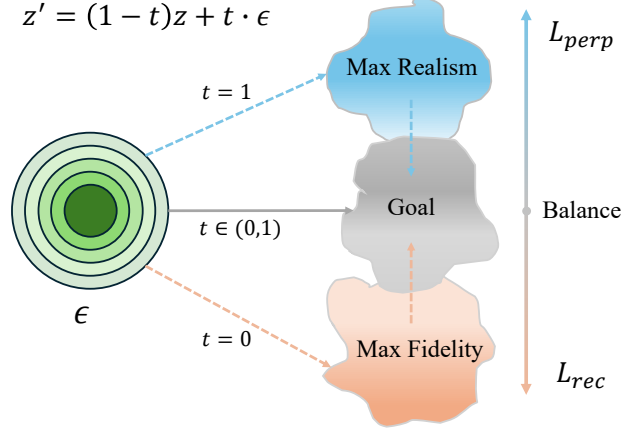


Figure 4. Illustration of fidelity-aware condition controller. To stabilize the restoration trajectory, flow-based noise augmentation and fidelity-aware timestep sampling are applied. The timestep  $t$  can control the trade-off between realism and fidelity.

### Training Objective

During training, the wavelet-enhanced encoder first transfer input frames  $X$  into latent variable  $z$ . Noise augmentation produces a perturbed latent  $\tilde{z}_t$ , and the DiT predicts the clean latent  $\hat{z}_t$  under the flow-matching objective. The overall loss combines a rectified flow term and a reconstruction term through the frozen decoder:

$$\mathcal{L} = \lambda_{\text{flow}} \|v_{\theta}(\hat{z}_t, t) - (\epsilon - X_{HR})\|_2^2 + \lambda_{\text{rec}} \|D(\hat{z}_t) - X_{HR}\|_2^2, \quad (5)$$

where  $\epsilon$  is randomly sampled from standard normal distribution, and  $D$  is the frozen decoder. The combination of flow-matching supervision and decoder-based reconstruction encourages the model to leverage both generative priors and frequency-enhanced latent cues.

At inference time, the model starts from a random noise from standard normal distribution, progressively refining it toward a clean latent  $\hat{z}_t$ . The frozen decoder  $D$  then reconstructs the HR video.

### Two-Stage Training Strategy

To ensure stable adaptation, we adopt a two-stage training strategy. In the first stage, we freeze the entire DiT and decoder, training only the wavelet-enhanced encoder to match the latent distribution of the original encoder. This step prevents the decoder from receiving out-of-distribution latents. In the second stage, we finetune the DiT while freezing the VAE encoder and decoder, enabling the model to learn restoration-specific flow dynamics without disrupting the pretrained generative prior. This strategy significantly reduces training cost while preserving the strong spatial-temporal priors learned from large-scale video data. Furthermore, we employ a progressive training strategy and an

Table 1. Quantitative comparisons on both synthetic (SPMCS, UDM10, YouHQ40) and real-world (VideoLQ, RealVSR, AIGC30) VSR benchmarks. The best and second-best results are marked in red and blue, respectively.

| Dataset | Metric    | RealViFormer | Upscale-A-Video | MGLD-VSR | STAR   | SeedVR-7B | SeedVR2-7B | Ours   |
|---------|-----------|--------------|-----------------|----------|--------|-----------|------------|--------|
| SPMCS   | PSNR↑     | 24.185       | 22.12           | 23.41    | 22.58  | 20.78     | 22.90      | 23.82  |
|         | SSIM↑     | 0.663        | 0.562           | 0.633    | 0.609  | 0.575     | 0.638      | 0.673  |
|         | LPIPS↓    | 0.378        | 0.405           | 0.369    | 0.420  | 0.395     | 0.322      | 0.294  |
|         | DISTS↓    | 0.186        | 0.198           | 0.166    | 0.229  | 0.166     | 0.134      | 0.127  |
| UDM10   | PSNR↑     | 26.70        | 23.29           | 26.11    | 24.66  | 24.29     | 26.26      | 28.10  |
|         | SSIM↑     | 0.796        | 0.717           | 0.772    | 0.747  | 0.731     | 0.798      | 0.845  |
|         | LPIPS↓    | 0.285        | 0.301           | 0.273    | 0.359  | 0.264     | 0.203      | 0.130  |
|         | DISTS↓    | 0.166        | 0.175           | 0.144    | 0.195  | 0.124     | 0.101      | 0.070  |
| YouHQ40 | PSNR↑     | 23.26        | 22.41           | 22.62    | 22.15  | 20.60     | 22.46      | 23.67  |
|         | SSIM↑     | 0.606        | 0.565           | 0.576    | 0.575  | 0.546     | 0.600      | 0.612  |
|         | LPIPS↓    | 0.362        | 0.375           | 0.356    | 0.451  | 0.323     | 0.274      | 0.237  |
|         | DISTS↓    | 0.193        | 0.186           | 0.166    | 0.213  | 0.134     | 0.110      | 0.104  |
| VideoLQ | NIQE↓     | 4.153        | 3.676           | 3.864    | 5.915  | 4.933     | 4.948      | 4.412  |
|         | MUSIQ↑    | 54.65        | 51.45           | 53.49    | 40.50  | 48.35     | 45.76      | 54.13  |
|         | CLIP-IQA↑ | 0.411        | 0.327           | 0.333    | 0.243  | 0.258     | 0.257      | 0.356  |
|         | DOVER↑    | 7.035        | 6.935           | 8.109    | 6.891  | 7.416     | 7.236      | 8.234  |
| RealVSR | NIQE↓     | 4.056        | 3.849           | 3.627    | 3.927  | 3.851     | 3.877      | 3.792  |
|         | MUSIQ↑    | 47.542       | 72.858          | 73.241   | 73.912 | 73.825    | 73.540     | 75.690 |
|         | CLIP-IQA↑ | 0.458        | 0.540           | 0.553    | 0.537  | 0.566     | 0.562      | 0.568  |
|         | DOVER↑    | 10.39        | 12.58           | 12.23    | 12.46  | 13.29     | 12.34      | 14.07  |
| AIGC30  | NIQE↓     | 4.618        | 3.924           | 3.917    | 4.449  | 3.809     | 4.554      | 3.325  |
|         | MUSIQ↑    | 49.890       | 54.318          | 53.863   | 52.170 | 56.921    | 50.055     | 60.092 |
|         | CLIP-IQA↑ | 0.298        | 0.382           | 0.377    | 0.326  | 0.411     | 0.300      | 0.414  |
|         | DOVER↑    | 7.318        | 8.176           | 8.128    | 7.492  | 8.202     | 7.485      | 8.731  |

image-video mixed training strategy to facilitate the training of the diffusion transformer. Specifically, we begin training with small patches and short clips to adapt the video diffusion transformer to the VSR task. Then, we fine-tune the base model on larger patches and longer clips. Additionally, we train on joint image-video data to mitigate the issue of insufficient video data quality.

## 4. Experiments

### 4.1. Implementation Details

We train WEVSR in two stages to ensure stable adaptation of the pretrained video generative model. In the first stage, we freeze the entire decoder and DiT and train only the multi-level DWT front-end for 50k iterations. This step aligns the wavelet-enhanced encoder output with the latent distribution expected by the pretrained decoder, preventing out-of-distribution artifacts. In the second stage, we unfreeze the DiT while keeping the VAE frozen, and train the full restoration pipeline for 300k iterations.

We use AdamW with a learning rate of  $1 \times 10^{-4}$  for the wavelet front-end and  $5 \times 10^{-5}$  for the DiT. A cosine decay scheduler is applied. The batch size is 16, each containing 8 consecutive frames. In the first stage, random cropping to  $256 \times 256$  is used during training. In the second stage, we random crop patches from  $512 \times 512$  resolution to  $768 \times 768$  resolution and employ from 13 to 37 frames for training.

### 4.2. Datasets

The training dataset consists of the subset of OpenVid-1M (Nan et al., 2024) and YouHQ (Zhou et al., 2024a). LR data are synthesized following the degradation pipeline in Real-ESRGAN (Wang et al., 2021). To train the VAE encoder, we mix HR and LR images and crop to the same size, forming a customized dataset.

We evaluate the VSR performance on both synthetic and real-world benchmarks. The synthetic LR videos are generated using the same degradation settings as training, on three datasets: SPMCS (Tao et al., 2017), UDM10 (Yi et al., 2019), and YouHQ40 (Zhou et al., 2024a). We select VideoLQ (Chan et al., 2022b), RealVSR (Yang et al., 2021), and AIGC30 (self-collected 30 AI-generated videos) as the real-world testing data.

### 4.3. Evaluation Metrics

We employ a representative set of quality metrics to evaluate our method. For distortion-oriented fidelity, we report PSNR (Huynh-Thu & Ghanbari, 2008) and SSIM (Wang et al., 2004). For perceptual realism, we include LPIPS (Zhang et al., 2018) and DISTS (Ding et al., 2020). No-reference measures such as NIQE (Mittal et al., 2012) and MUSIQ (Ke et al., 2021) assess naturalness and overall perceptual quality. To capture semantic and temporal consistency, we adopt CLIP-IQA (Wang et al., 2023), DOVER (Wu et al., 2023), and rFVD (Unterthiner et al.,

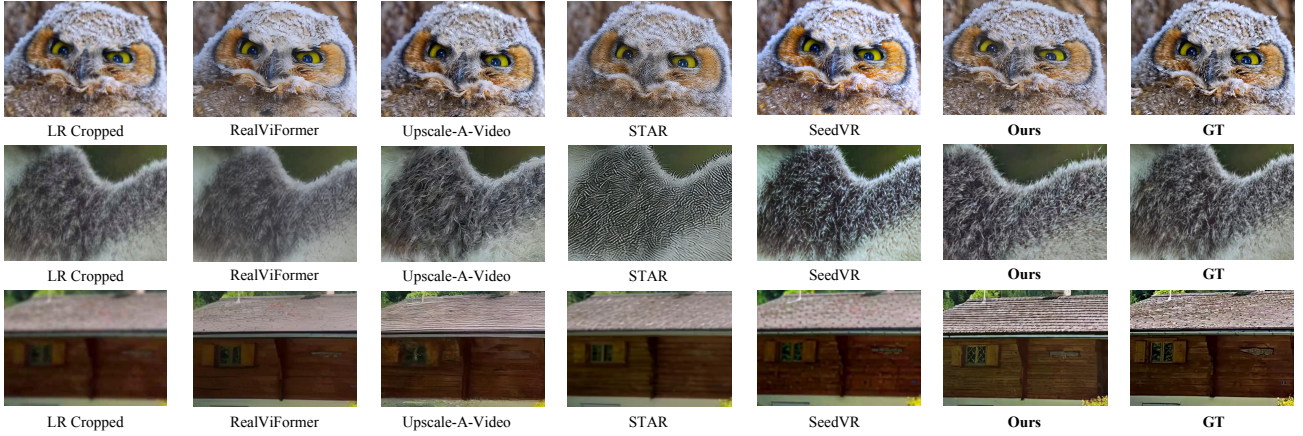


Figure 5. Quantitative comparisons on super-resolved video frames from synthetic datasets. Compared to the state-of-the-art VSR models (Zhang & Yao, 2024; Zhou et al., 2024a; Xie et al., 2025b; Wang et al., 2025a), our results demonstrate more natural facial details and better structure of the text. (Zoom-in for best view)



Figure 6. Quantitative comparisons on super-resolved video frames from real-world datasets. Compared to the state-of-the-art VSR models (Zhang & Yao, 2024; Zhou et al., 2024a; Xie et al., 2025b; Wang et al., 2025a), our results demonstrate more natural facial details and better structure of the text. (Zoom-in for best view)

2018). Together, these metrics provide a balanced evaluation across fidelity, perceptual quality, and temporal coherence.

#### 4.4. VSR Comparisons with State-of-the-Art

**Quantitative comparisons.** We compare our approach with recent state-of-the-art RealVSR methods, including **Transformer-based** method RealViFormer (Zhang & Yao, 2024), and **Diffusion-based** approaches such as Upscale-A-Video (Zhou et al., 2024a), MGLD-VSR (Yang et al., 2024a), STAR (Xie et al., 2025b), SeedVR (Wang et al., 2025b), and SeedVR2 (Wang et al., 2025a).

On synthetic datasets such as SPMCS, UDM10, and YouHQ40, our approach delivers competitive PSNR while significantly outperforming all baselines in SSIM, LPIPS,

and DISTS. For instance, our model achieves the best SSIM, LPIPS and DISTS on all three datasets, indicating more faithful texture reconstruction compared to RealViFormer, Upscale-A-Video, MGLD-VSR, STAR, and SeedVR. For PSNR, our method also attains the best performance on UDM10 and YouHQ40, second best for SPMCS, demonstrating strong fidelity under various circumstance.

The improvements become less pronounced on VideoLQ, which contains more challenging real-world degradations that are hard to handle for diffusion models. While traditional transformer-based method RealViFormer achieves the best MUSIQ and CLIP-IQA, our model achieves the best MUSIQ and CLIP-IQA among all diffusion-based methods, indicating superior semantic alignment. Furthermore, our model achieves the best DOVER, demonstrating its excel-



Table 2. Quantitative comparisons of different VAE methods on video reconstruction benchmark. The best and second-best results are marked in red and blue, respectively.

| Methods (VAE) | Params (M) | Compression | Spatial | Latent | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | rFVD $\downarrow$ |
|---------------|------------|-------------|---------|--------|-----------------|-----------------|--------------------|-------------------|
| OpenSora      | 393        | 4           | 8       | 4      | 26.32           | 0.8448          | 0.1578             | 44.69             |
| OpenSoraPlan  | 180        | 4           | 8       | 16     | 28.89           | 0.8816          | 0.0927             | 26.32             |
| Cosmos        | 90         | 4           | 8       | 16     | 30.72           | 0.9010          | 0.0805             | 12.37             |
| CV-VAE (SD3)  | 182        | 4           | 8       | 16     | 31.55           | 0.9131          | 0.0560             | 6.18              |
| CogVideoX     | 217        | 4           | 8       | 16     | 32.59           | 0.9168          | 0.0592             | 5.76              |
| SeedVR        | 251        | 4           | 8       | 16     | 32.14           | 0.9161          | 0.0491             | 1.76              |
| WAN           | 127        | 4           | 8       | 16     | 33.40           | 0.9189          | 0.0514             | 3.38              |
| Ours          | 127        | 4           | 8       | 16     | 33.65           | 0.9203          | 0.0497             | 2.96              |

lent temporal consistency.

On the other real-world benchmarks, our approach establishes a clear margin over all competing methods. It achieves the second best NIQE, reflecting its ability to recover sharper and more natural textures. Moreover, our method maintains best performance in MUSIQ, CLIP-IQA, and DOVER, further validating its robustness and temporal consistency under real-world degradations where handcrafted degradations fail to approximate the true distribution.

Overall, the quantitative results demonstrate that our method consistently surpasses both regression-based and diffusion-based baselines across all datasets. The substantial gains in perceptual metrics highlight the effectiveness of our wavelet-enhanced latent modeling and diffusion refinement strategy, enabling high-fidelity and temporally stable video super-resolution in real-world scenarios.

**Qualitative comparisons.** To further evaluate the perceptual quality of our method, we present visual comparisons against recent state-of-the-art approaches, including RealViFormer (Zhang & Yao, 2024), Upscale-A-Video (Zhou et al., 2024a), STAR (Xie et al., 2025b), and SeedVR/SeedVR2 (Wang et al., 2025a), on representative samples from the YouHQ (Zhou et al., 2024a) and VideoLQ (Chan et al., 2022b) datasets. As shown in Figure 6, existing methods struggle to simultaneously recover fine-grained textures and maintain structural fidelity under real-world degradations. Approaches such as STAR and Upscale-A-Video tend to oversmooth high-frequency regions, leading to blurry edges and loss of material details. RealViFormer and SeedVR preserve sharper boundaries but often introduce ringing artifacts or inconsistent textures, especially in challenging regions such as thin structures, repetitive patterns, and low-light areas.

In contrast, our method reconstructs significantly more faithful details while avoiding hallucinated patterns. The wavelet-enhanced encoder provides stronger frequency decomposition, enabling accurate recovery of subtle textures that other methods fail to preserve. Meanwhile, the noise-guided diffusion refinement effectively suppresses

noise and artifacts, producing cleaner and more stable outputs. Across all examples, our results exhibit sharper contours, more natural textures, and fewer flickering artifacts, closely matching the realistic appearance.

#### 4.5. VAE Comparisons with State-of-the-Art

We compare our VAE with a wide range of state-of-the-art latent video VAEs, including OpenSora (Zheng et al., 2024), OpenSoraPlan (Lab & etc., 2024), Cosmos (Reda et al., 2024), CV-VAE (SD3) (Zhao et al., 2024), CogVideoX (Yang et al., 2024b), SeedVR (Wang et al., 2025b), and WAN (Wan et al., 2025). As shown in Table 2, our model achieves the best overall performance across both distortion-oriented and perceptual metrics while maintaining the same latent configuration (temporal compression  $4\times$ , spatial compression  $8\times$ , and 16 latent channels) as recent high-capacity VAEs.

In terms of reconstruction fidelity, our VAE attains the highest PSNR and SSIM, surpassing WAN and significantly outperforming CogVideoX and SD3-VAE. These improvements indicate that our wavelet-enhanced latent representation preserves fine-grained spatial structures more effectively than existing diffusion-oriented VAEs.

Our method also demonstrates strong perceptual advantages. It achieves a second best LPIPS, reflecting more faithful texture reconstruction and reduced perceptual distortion. Notably, our VAE delivers competitive performance relative to SeedVR, despite using substantially fewer parameters (127M vs. 251M), highlighting the efficiency of our design. The improvements are even more pronounced in temporal perceptual quality. Our model achieves a second best rFVD, outperforming WAN and approaching SeedVR, which benefits from adversarial post-training and a significantly larger model size. This demonstrates that our latent modeling strategy effectively captures temporal dynamics without relying on heavy adversarial tuning.

Overall, the quantitative results show that our VAE establishes a new balance between reconstruction fidelity, perceptual realism, and model efficiency. It consistently out-

Table 3. Ablation study on various techniques. The best and second-best results are marked in red and blue, respectively.

| Timestep | Noise | Wavelet | PSNR↑ | SSIM↑ | LPIPS↓ | DISTS↓ |
|----------|-------|---------|-------|-------|--------|--------|
|          |       |         | 22.54 | 0.718 | 0.233  | 0.111  |
| ✓        |       |         | 23.82 | 0.754 | 0.194  | 0.097  |
|          | ✓     |         | 24.65 | 0.783 | 0.175  | 0.087  |
|          |       | ✓       | 25.43 | 0.837 | 0.168  | 0.092  |
| ✓        | ✓     |         | 25.96 | 0.812 | 0.163  | 0.075  |
| ✓        |       | ✓       | 26.74 | 0.828 | 0.152  | 0.083  |
|          | ✓     | ✓       | 27.57 | 0.839 | 0.143  | 0.079  |
| ✓        | ✓     | ✓       | 28.10 | 0.845 | 0.130  | 0.070  |

Table 4. Efficiency comparison on different VSR models. The best and second-best results are marked in red and blue, respectively.

| Metrics        | UAV  | MGLD | STAR | SeedVR | SeedVR2 | Ours |
|----------------|------|------|------|--------|---------|------|
| Parameters (M) | 691  | 1431 | 2041 | 8240   | 8240    | 1546 |
| VRAM (GB)      | 64   | 36   | 64   | 72     | 72      | 24   |
| Infer time (s) | 1213 | 1276 | 2145 | 1362   | 286     | 757  |

performs prior VAEs across all key metrics, confirming the effectiveness of our wavelet-enhanced latent encoding and its suitability as a foundation for high-quality video super-resolution and generation.

#### 4.6. Ablation Studies

To assess the contribution of each proposed component, we conduct a systematic ablation study in Table 3. Starting from the baseline without any modules, we progressively introduce Timestep Sampling, Noise Augmentation, and the Wavelet Encoder. Each technique yields consistent improvements across all metrics, confirming their effectiveness in enhancing fidelity and perceptual quality.

Individually, all three techniques improve both the fidelity and perception metrics compared to the baseline. The wavelet-enhanced encoder provides the most significant single-component improvement. When combined, these modules exhibit strong complementarity. Pairing timestep sampling with either noise augmentation or the wavelet encoder further boosts reconstruction quality, and the joint use of noise augmentation and wavelet priors achieves the second best performance on PSNR, SSIM, and LPIPS. Ultimately, integrating all three techniques delivers the best overall performance, demonstrating that the synergy is crucial for achieving state-of-the-art performance.

Visual results in Figure 7 further illustrate that as the noise level increases, the generative capacity is enhanced but the fidelity is compromised, highlighting the importance of controlling perturbations by giving more weight to intermediate timesteps via sampling them more frequently.

#### 4.7. Efficiency Analysis

Table 4 summarizes the efficiency of recent VSR models. While large-scale generative baselines such as SeedVR and

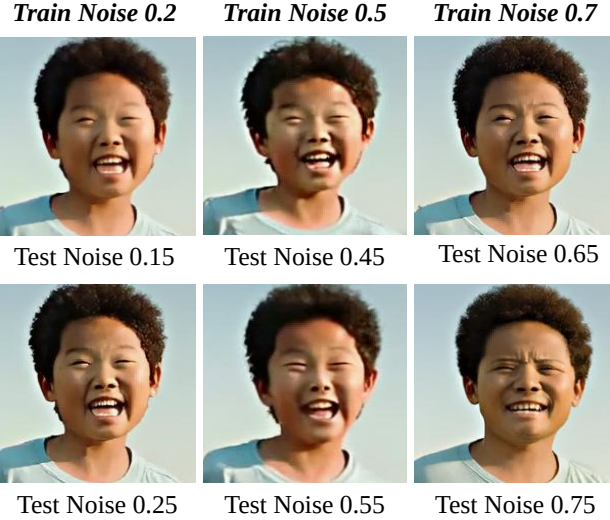


Figure 7. Visual results of WEVSR trained and tested under varying noise levels. Training with a high noise level demonstrates generative capacity but low fidelity. Additionally, introducing high-level noise increases training difficulty and leads to suboptimal outcomes. (Zoom in for best view)

SeedVR2 produce great results, they require massive parameters and extensive VRAM usage. In contrast, our model achieves a balanced design: with only 1.5B parameters and 24 GB memory consumption. Moreover, SeedVR2 achieves the fastest inference time at the cost of 72GB memory. Our approach delivers a favorable trade-off, offering substantially lower memory requirements and competitive inference speed, thereby demonstrating practical scalability for real-world deployment.

## 5. Limitation

Despite strong performance on various benchmarks, our approach still faces challenges under extreme degradations and highly dynamic scenes. The reliance on synthetic training data may limit adaptability to unseen distortions, and real-time performance remains constrained compared to specialized streaming models.

## 6. Conclusion

We introduced a VSR framework building upon a pretrained video generation diffusion transformer. Integrating timestep sampling, noise augmentation, and frequency priors, our method reshapes the flow-matching dynamics to better preserve structural stability while maintaining fine-detail recovery. Ablation studies demonstrate that each component contributes meaningfully, and their synergy yields state-of-the-art performance. Quantitative and qualitative evaluation on multiple synthetic and real-world benchmarks demonstrate the generalization ability of our approach, which establishes a strong foundation for future research.



## Acknowledgements

This work was supported by the National Natural Science Foundation of China (Grant No. 62322216, Grant No. U24B20175), Scientific Research Innovation Capability Support Project for Young Faculty (No.SRICSPYF-ZY2025012), and Shenzhen Science and Technology Program (No.SYSRD20250529113401002). This research was also financially supported by Tencent Technology (Grant No. JR2025TEG013) and the Open Research Fund from Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), under Grant No. GML-KF-26-02.

## Impact Statement

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

## References

- Caballero, J., Ledig, C., Aitken, A., Acosta, A., Totz, J., Wang, Z., and Shi, W. Real-time video super-resolution with spatio-temporal networks and motion compensation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4778–4787, 2017.
- Chan, K. C., Wang, X., Yu, K., Dong, C., and Loy, C. C. BasicVSR: The search for essential components in video super-resolution and beyond. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- Chan, K. C., Zhou, S., Xu, X., and Loy, C. C. BasicVSR++: Improving video super-resolution with enhanced propagation and alignment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022a.
- Chan, K. C., Zhou, S., Xu, X., and Loy, C. C. Investigating tradeoffs in real-world video super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022b.
- Chen, Z., Zou, Z., Zhang, K., Su, X., Yuan, X., Guo, Y., and Zhang, Y. Dove: Efficient one-step diffusion model for real-world video super-resolution. *arXiv preprint arXiv:2505.16239*, 2025.
- Ding, K., Ma, K., Wang, S., and Simoncelli, E. P. Image quality assessment: Unifying structure and texture similarity. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2020.
- Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Proceedings of International Conference on Machine Learning (ICML)*, 2024a.
- Esser, P., Kulal, S., Blattmann, A., Entezari, R., Müller, J., Saini, H., Levi, Y., Lorenz, D., Sauer, A., Boesel, F., et al. Scaling rectified flow transformers for high-resolution image synthesis. In *Forty-first international conference on machine learning*, 2024b.
- HaCohen, Y., Chiprut, N., Brazowski, B., Shalem, D., Moshe, D., Richardson, E., Levin, E., Shiran, G., Zabari, N., Gordon, O., et al. Ltx-video: Realtime video latent diffusion. *arXiv preprint arXiv:2501.00103*, 2024.
- Huynh-Thu, Q. and Ghanbari, M. Scope of validity of psnr in image/video quality assessment. *Electronics letters*, 44 (13):800–801, 2008.
- Jo, Y., Wug Oh, S., Kang, J., and Joo Kim, S. Deep video super-resolution network using dynamic upsampling filters without explicit motion compensation. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, pp. 3224–3232, 2018.
- Ke, J., Wang, Q., Wang, Y., Milanfar, P., and Yang, F. Musiq: Multi-scale image quality transformer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- Lab, P.-Y. and etc., T. A. Open-sora-plan. <https://doi.org/10.5281/zenodo.10948109>, 2024.
- Lai, B., Wang, X., Rambhatla, S., Rehg, J. M., Kira, Z., Girdhar, R., and Misra, I. Toward diffusible high-dimensional latent spaces: A frequency perspective. *arXiv preprint arXiv:2511.22249*, 2025.
- Li, W., Tao, X., Guo, T., Qi, L., Lu, J., and Jia, J. Mucan: Multi-correspondence aggregation network for video super-resolution. *arXiv preprint arXiv:2007.11803*, 2020.
- Li, X., Liu, Y., Cao, S., Chen, Z., Zhuang, S., Chen, X., He, Y., Wang, Y., and Qiao, Y. Diffvsr: Revealing an effective recipe for taming robust video super-resolution against complex degradations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15319–15328, 2025.
- Liang, J., Fan, Y., Xiang, X., Ranjan, R., Ilg, E., Green, S., Cao, J., Zhang, K., Timofte, R., and Gool, L. V. Recurrent video restoration transformer with guided deformable attention. *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

- Lipman, Y., Chen, R. T., Ben-Hamu, H., Nickel, M., and Le, M. Flow matching for generative modeling. *arXiv preprint arXiv:2210.02747*, 2022.
- Liu, X., Gong, C., and Liu, Q. Flow straight and fast: Learning to generate and transfer data with rectified flow. *arXiv preprint arXiv:2209.03003*, 2022.
- Mittal, A., Soundararajan, R., and Bovik, A. C. Making a “completely blind” image quality analyzer. *IEEE Signal processing letters*, 2012.
- Nan, K., Xie, R., Zhou, P., Fan, T., Yang, Z., Chen, Z., Li, X., Yang, J., and Tai, Y. Openvid-1m: A large-scale high-quality dataset for text-to-video generation. *arXiv preprint arXiv:2407.02371*, 2024.
- Peebles, W. and Xie, S. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 4195–4205, 2023.
- Reda, F., Gu, J., Liu, X., Ge, S., Wang, T.-C., Wang, H., and Liu, M.-Y. Cosmos tokenizer: A suite of image and video neural tokenizers. <https://github.com/NVIDIA/Cosmos-Tokenizer>, 2024.
- Sun, Y., Sun, L., Liu, S., Wu, R., Zhang, Z., and Zhang, L. One-step diffusion for detail-rich and temporally consistent video super-resolution. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- Tao, X., Gao, H., Liao, R., Wang, J., and Jia, J. Detail-revealing deep video super-resolution. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2017.
- Unterthiner, T., Van Steenkiste, S., Kurach, K., Marinier, R., Michalski, M., and Gelly, S. Towards accurate generative models of video: A new metric & challenges. *arXiv preprint arXiv:1812.01717*, 2018.
- Wan, T., Wang, A., Ai, B., Wen, B., Mao, C., Xie, C.-W., Chen, D., Yu, F., Zhao, H., Yang, J., Zeng, J., Wang, J., Zhang, J., Zhou, J., Wang, J., Chen, J., Zhu, K., Zhao, K., Yan, K., Huang, L., Feng, M., Zhang, N., Li, P., Wu, P., Chu, R., Feng, R., Zhang, S., Sun, S., Fang, T., Wang, T., Gui, T., Weng, T., Shen, T., Lin, W., Wang, W., Wang, W., Zhou, W., Wang, W., Shen, W., Yu, W., Shi, X., Huang, X., Xu, X., Kou, Y., Lv, Y., Li, Y., Liu, Y., Wang, Y., Zhang, Y., Huang, Y., Li, Y., Wu, Y., Liu, Y., Pan, Y., Zheng, Y., Hong, Y., Shi, Y., Feng, Y., Jiang, Z., Han, Z., Wu, Z.-F., and Liu, Z. Wan: Open and advanced large-scale video generative models. *arXiv preprint arXiv:2503.20314*, 2025.
- Wang, J., Chan, K. C., and Loy, C. C. Exploring clip for assessing the look and feel of images. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023.
- Wang, J., Lin, S., Lin, Z., Ren, Y., Wei, M., Yue, Z., Zhou, S., Chen, H., Zhao, Y., Yang, C., Xiao, X., Loy, C. C., and Jiang, L. Seedvr2: One-step video restoration via diffusion adversarial post-training, 2025a. URL <https://arxiv.org/abs/2506.05301>.
- Wang, J., Lin, Z., Wei, M., Zhao, Y., Yang, C., Loy, C. C., and Jiang, L. Seedvr: Seeding infinity in diffusion transformer towards generic video restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025b.
- Wang, X., Chan, K. C., Yu, K., Dong, C., and Change Loy, C. Edvr: Video restoration with enhanced deformable convolutional networks. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (CVPR-W)*, 2019.
- Wang, X., Xie, L., Dong, C., and Shan, Y. Real-esrgan: Training real-world blind super-resolution with pure synthetic data. In *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops (ICCV-W)*, 2021.
- Wang, Z., Bovik, A. C., Sheikh, H. R., and Simoncelli, E. P. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing (TIP)*, 2004.
- Wang, Z., Zhao, G., Ren, J., Feng, B., Zhang, S., and Li, W. Turbovsr: Fantastic video upscalers and where to find them. *arXiv preprint arXiv:2506.23618*, 2025c.
- Wu, H., Zhang, E., Liao, L., Chen, C., Hou, J., Wang, A., Sun, W., Yan, Q., and Lin, W. Exploring video quality assessment on user generated contents from aesthetic and technical perspectives. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.
- Xie, L., Li, Y., Du, S., Xia, M., Wang, X., Yu, F., Chen, Z., Wan, P., Zhou, J., and Dong, C. Simplegvr: A simple baseline for latent-cascaded video super-resolution. *arXiv preprint arXiv:2506.19838*, 2025a.
- Xie, R., Liu, Y., Zhou, P., Zhao, C., Zhou, J., Zhang, K., Zhang, Z., Yang, J., Yang, Z., and Tai, Y. Star: Spatial-temporal augmentation with text-to-video models for real-world video super-resolution. *arXiv preprint arXiv:2501.02976*, 2025b.
- Xu, K., Yu, Z., Wang, X., Mi, M. B., and Yao, A. Enhancing video super-resolution via implicit resampling-based alignment. In *2024 IEEE/CVF Conference on Computer*

- Vision and Pattern Recognition (CVPR)*, pp. 2546–2555, 2024. doi: 10.1109/CVPR52733.2024.00246.
- Yang, X., Xiang, W., Zeng, H., and Zhang, L. Real-world video super-resolution: A benchmark dataset and a decomposition based learning scheme. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.
- Yang, X., He, C., Ma, J., and Zhang, L. Motion-guided latent diffusion for temporally consistent real-world video super-resolution. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024a.
- Yang, Z., Teng, J., Zheng, W., Ding, M., Huang, S., Xu, J., Yang, Y., Hong, W., Zhang, X., Feng, G., et al. Cogvideox: Text-to-video diffusion models with an expert transformer. *arXiv preprint arXiv:2408.06072*, 2024b.
- Yi, P., Wang, Z., Jiang, K., Jiang, J., and Ma, J. Progressive fusion video super-resolution network via exploiting non-local spatio-temporal correlations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.
- Yu, L., Lezama, J., Gundavarapu, N. B., Versari, L., Sohn, K., Minnen, D., Cheng, Y., Gupta, A., Gu, X., Hauptmann, A. G., Gong, B., Yang, M.-H., Essa, I., Ross, D. A., and Jiang, L. Language model beats diffusion - tokenizer is key to visual generation. In *Proceedings of International Conference on Learning Representations (ICLR)*, 2024.
- Zhang, B. and Sennrich, R. Root mean square layer normalization. *Advances in neural information processing systems*, 32, 2019.
- Zhang, R., Isola, P., Efros, A. A., Shechtman, E., and Wang, O. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018.
- Zhang, Y. and Yao, A. RealViformer: Investigating attention for real-world video super-resolution. *Proceedings of the European Conference on Computer Vision (ECCV)*, 2024.
- Zhao, S., Zhang, Y., Cun, X., Yang, S., Niu, M., Li, X., Hu, W., and Shan, Y. Cv-vae: A compatible video vae for latent generative video models. *Proceedings of Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Zhao, Y., Su, Y.-C., Chu, C.-T., Li, Y., Renn, M., Zhu, Y., Chen, C., and Jia, X. Rethinking deep face restoration. In *cvpr*, 2022.
- Zheng, Z., Peng, X., Yang, T., Shen, C., Li, S., Liu, H., Zhou, Y., Li, T., and You, Y. Open-sora: Democratizing efficient video production for all. <https://github.com/hpcaitech/Open-Sora>, 2024.
- Zhou, S., Yang, P., Wang, J., Luo, Y., and Loy, C. C. Upscale-A-Video: Temporal-consistent diffusion model for real-world video super-resolution. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024a.
- Zhou, X., Zhang, L., Zhao, X., Wang, K., Li, L., and Gu, S. Video super-resolution transformer with masked inter&intra-frame attention. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 25399–25408, 2024b.
- Zhuang, J., Guo, S., Cai, X., Li, X., Liu, Y., Yuan, C., and Xue, T. Flashvsr: Towards real-time diffusion-based streaming video super-resolution. *arXiv preprint arXiv:2510.12747*, 2025.